

Background: Understand Needs

Highlights

- A true story: Voting trouble experienced by a senior citizen.
- History of contextual inquiry:
 - Roots in activity theory.
 - Roots in ethnography.
 - Getting contextual studies into HCI.
 - Connections to participatory design.
- The SSA Model District Office—An extreme and successful example of an environment for data elicitation.
- Roots of essential use cases.

11.1 THIS IS A REFERENCE CHAPTER

This chapter contains reference material relevant to the other chapters of [Part 2](#). This chapter is not intended to be read through as a regular chapter, but each section is supposed to be read when a reference to it in the main chapters is encountered.

11.2 A TRUE STORY: VOTING TROUBLE EXPERIENCED BY A SENIOR CITIZEN

In southwest Virginia, somewhat remote from urban centers, when the first-time computer-based touchscreen voting machines were used, we heard that quite a few voters had difficulty in using them. Although an official gave instructions as people entered one particular voting area, a school gymnasium, he did it in a confusing way.

One of the voters in line was an elderly woman with poor eyesight, obvious from her thick eyeglasses. As she entered the voting booth, one could just imagine her leaning her head down very close to the screen, struggling to read the words, even though the font was reasonably large.

Her voice was heard floating above her voting booth, as she gave some unsolicited user feedback. She was saying that she had trouble distinguishing the colors (the screen was in primary colors: red, green, and blue). A bystander said aloud to himself, as if to answer the woman, that he thought there was an option to set the screen to black and white. But oddly, no one actually told this, if it was true, to the woman.

In time, the woman emerged with a huge smile, proclaiming victory over the evil machine. She then immediately wanted to tell everyone how the design should be improved. Remember, this is an elderly woman who probably knew nothing about technology or user experience but who is quite naturally willing to offer valuable user feedback.

It was easy to imagine a scenario in which the supervisors of the voting process quickly flocked about the voter and duly took notes, pledging to pass this important information on to the higher-ups who could influence the next design. But as you might guess, she was roundly humored and ignored.

There are a few things to note in this story. First, the feedback was rich, detailed, and informative about design. This level of feedback was possible only because it occurred in real usage and real context.

Second, this woman represented a particular type of user belonging to a specific age group and had some associated visual limitations. She was also naturally articulate in describing her usage experience, which is somewhat uncommon in public situations.

So what does this have to do with user research inquiry? If you do user research inquiry in a real environment like this, you might get lucky and find rich user data. It is certain, however, that if you do not do user research inquiry, you will never get this kind of information about situated usage.

11.3 HISTORY OF CONTEXTUAL INQUIRY

The terms contextual inquiry, contextual design, and contextual studies are early terms for what we are now calling usage research.

11.3.1 Roots in Activity Theory

First of all, we owe a great acknowledgment to those who pioneered, developed, and promoted the concepts and techniques of contextual design. Early foundations go back to Scandinavian work activity theory (Bjerknes, Ehn, & Kyng, 1987; Bødker, 1991; Ehn, 1988).

In our context, activity theory is an abstract adaptation for HCI of a theoretical descriptive framework based on the Soviet psychological activity theory as adapted by Scandinavian researchers in the mid-1980s.

The goal seems to have been to understand people (users of the systems) as complex human beings and not information processors or system components. People's actions are influenced by their social and cultural setting. This theory is based on learning what motivates people to take actions. Human activities, especially in the context of interacting with machines and systems, are seen as social actions, taking into account the individual characteristics and abilities of the human operators.

The activity theory work was conducted for quite some time in Scandinavia, in parallel with the task analysis work in Europe and the United Kingdom. More recent conferences and special issues have been devoted to the topic (Lantz & Gulliksen, 2003). Much of the initial work in this "school" was directed at the impact of computer-based systems on human labor and democracy within the organizations of the affected workers. This singular focus on human work activities shines through into contextual inquiry and analysis.

11.3.2 Roots in Ethnography

A second essential foundation for contextual inquiry is ethnography, an investigative field rooted in anthropology (LeCompte & Preissle, 1993). Anthropologists spend significant amounts of time living with and studying a particular group of humans or other possibly more intelligent animals, usually in social settings of primitive cultures. The goal is to study and document details of their daily lives and existence.

The Understand Needs lifecycle activity in UX derives from ethnography, a branch of anthropology that focuses on the study and systematic description of various human cultures.

In a trend toward design driven by work practice in context, quick and dirty varieties of ethnography, along with other hermeneutic approaches (concerned with ways to explain, translate, and interpret perceived reality) (Carroll, Mack, & Kellogg, 1988), have been adapted into HCI practice as qualitative tools for understanding design requirements. Contextual inquiry and analysis are examples of an adaptation of this kind of approach as part of the evolution of requirement elicitation techniques.

The characteristics that define ethnography in anthropology are what make it just right for adaptation in HCI, where it takes place in the natural setting of the people being studied. It also involves observation of user activities, listening to what users say, asking questions, and discussing the work with the people who do it; and it is based on the holistic view of understanding behavior in its context.

In contrast to long-term field studies of “pure” ethnography, with its cultural, anthropological, and social perspectives, the “quick and dirty” version of ethnography has been adapted for HCI. Although involving significantly shorter time with subjects and correspondingly less depth of analysis, this version still requires observation of subjects in their own environment and still requires attending to the sociality of the subjects in their work context (Hughes, King, Rodden, & Andersen, 1995). For example, Hughes, King, Rodden, and Andersen (1994) describe the application of ethnography in the area of computer-supported cooperative work (CSCW), a subarea of HCI.

Lewis, Mateas, Palmiter, and Lynch (1996) described an ethnographic-based approach to system requirements and design that parallels much of the contextual inquiry process described here. Rogers and Bellotti (1997) tell how they harnessed ethnography as a research tool to serve as a practical requirements and design process. Blythin, Rouncefield, and Hughes (1997) address the adaptation of ethnography from research to commercial system development.

11.3.3 Getting Contextual Studies into HCI

The foundations for contextual design in HCI were laid by researchers at Digital Equipment Corporation (Whiteside & Wixon, 1987; Wixon, 1995; Wixon, Holtzblatt, & Knox, 1990). By 1988, several groups in academia and industry were already reporting on early contextual field studies (Good, 1989) in the United States and the United Kingdom (notably the work of Andrew Monk). Similar trends were also beginning in the software world (Suchman, 1987). Almost a decade later, Wixon and Ramey (1996) produced an edited collection of much more in-depth reports on case studies of real application of contextual studies in the field. Whiteside, Bennett, and Holtzblatt (1988) helped integrate the concept of contextual studies into the UX process.

11.3.4 Connections to Participatory Design

*Participatory design is a democratic process for UX design actively involving all stakeholders (e.g., employees, partners, customers, citizens, users) in the process to help ensure the result meets their needs and is usable. Underlying participatory design are the arguments that users should be involved in designs they will be using, and that all stakeholders, including and especially users, have equal inputs into UX design.*¹

¹https://en.wikipedia.org/wiki/Participatory_design.

Contextual inquiry and analysis are part of a collection of collaborative and participatory methods that evolved in parallel courses over the past couple of decades. These methods share the characteristic that they directly involve users not trained in specialized methods, such as task analysis. Among these are participatory design and collaborative analysis of requirements and design developed by Muller and associates (Muller & Kuhn, 1993; Muller, Wildman, & White, 1993a, 1993b) and collaborative users' task analysis (Lafrenière, 1996).

11.4 THE SSA MODEL DISTRICT OFFICE—AN EXTREME AND SUCCESSFUL EXAMPLE OF AN ENVIRONMENT FOR DATA ELICITATION

In the mid-1990s, we worked extensively with the Social Security Administration in Baltimore, mainly in usability engineering (what UX used to be called then) lifecycle training. We worked with a small group pioneering the introduction of usability engineering techniques into an “old school,” Waterfall-oriented, mainframe software development environment. Large Social Security systems were slowly migrating from mainframes (in Baltimore) plus terminals (by the thousands over the country) to client-server applications, some starting to run on PCs, and they wanted usability to be a high priority. Sean Wheeler was the group sparkplug and usability champion, strongly supported by Annette Bryce and Pat Stoos.

What impressed us the most about this organization was the Model District Office (MDO). A decade earlier, as part of a large Claims Modernization Project, a program of system design and training to “revolutionize the way that SSA serves the public,” they had built a complete and detailed model of a district office from middle America right in the middle of SSA headquarters in Baltimore. The MDO was indistinguishable from a typical agency office in a typical town, with its carpeting, office furniture, and computer terminals, right down to the office lamps and pictures on the wall. They brought in real SSA employees from field offices all over the United States to sit in the MDO to test and pilot new systems and procedures.

Although the MDO was originally intended for developing and refining the ways agents interact with clients, when SSA was ready to focus on usability, the existing MDO was the perfect environment for developing ways that agents interact with computer systems and for usability testing of those designs. Simply put, it was an extreme and successful example of leveraging ecological validity (the extent to which your UX evaluation setup matches the user's real work

context) for application development and testing as well as for user training. In the end, the group created a usability success story amid the inertia and enormous weight of the rest of the Social Security Administration. As a result of their work, this group won a federal award for quality and usability in a large and important software application.

As a testament to their seriousness about ecological validity and usability, by the mid- to late-1990s, the SSA was spending \$1 million a year to bring in employees to stay and work at the MDO, sometimes for a few months at a time. Their cost justification calculations proved the activity was saving many times more.

11.5 ROOTS OF ESSENTIAL USE CASES IN SOFTWARE ENGINEERING

A use case is not a user experience lifecycle artifact, but a software engineering and systems engineering artifact for documenting functional requirements, especially for object-oriented development, of a system. “Use cases, stated simply, allow description of sequences of events that, taken together, lead to a system doing something useful” (Bittner & Spence, 2003). They include outside roles—end users and external entities such as database servers or bank authorization modules—and internal system responses to outside actions.

Although a use case can represent the user view, the bottom-line focus is on functional, not interaction, requirements. Sometimes use cases are thought of as an object-oriented approach to user modeling, but in practice they are usually created by developers and systems analysts without any user research data from users.

Use cases are formalized usage scenarios, narratives of “black box” functionality in the context of user-system interaction (Constantine & Lockwood, 1999, p. 101). Use cases are often used as a component of software requirements. Their strong software orientation means that use cases lean in the direction of software implementation and away from user interaction design. As Meads (2010) says, in use cases, the user is an external object, not a person with human needs and limitations. This view leads to system requirements, but not to usage or UX requirements.

Use cases describe the major business requirements, features, and functions that the envisioned system must support. A use case “describes a sequence of actions that are performed by a human in work roles or other entities such as a machine or another system as they interact with the software” (Pressman, 2009);

“use cases help to identify the scope of the project and provide a basis for project planning” ([Pressman, 2009](#)).

In answer to the need for something more effective than use cases in identifying interaction design requirements, [Constantine \(1994a, 1995\)](#) created a variation he calls “essential use cases.”